

CS Notes (Groups)

Jesse Oldroyd

June 22, 2026

2026-06-22

CS Notes (Groups)

CS Notes (Groups)

Jesse Oldroyd
June 22, 2026

Introduction

Creating Groups in Python

2026-06-22

└ Outline

Introduction

Creating Groups in Python

2026-06-22

CS Notes (Groups)
└ Introduction

Introduction

Introduction

We will use **Python** in this course to perform mathematical computations. For the first part of the computer science course, we will look at how to use Python to work with groups. We will primarily interact with Python using the **Spyder** IDE.

2026-06-22

CS Notes (Groups)

└ Introduction

└ Python

Python

We will use **Python** in this course to perform mathematical computations. For the first part of the computer science course, we will look at how to use Python to work with groups. We will primarily interact with Python using the **Spyder** IDE.

The fundamental objects we need from Python are:

- **lists**: [obj1, obj2, ...]
- **functions**: def func(x): ...

The fundamental objects we need from Python are:

- **lists**: [obj1, obj2, ...]
- **functions**: def func(x): ...

2026-06-22

CS Notes (Groups)
└─ Creating Groups in Python

Creating Groups in Python

Creating Groups in Python

A Simple Group in Python

Suppose we want to model the coins group of the previous class. Then we will need to specify the *objects* and the *actions* of the group. We can use lists to hold the objects. . .

```
# group from 1.1 has as objects arrangements of a
# penny and a dime
arrangement1 = ['penny', 'nickel']
arrangement2 = ['nickel', 'penny']
objects = [arrangement1, arrangement2]
```

2026-06-22

CS Notes (Groups)

└ Creating Groups in Python

└ A Simple Group in Python

Suppose we want to model the coins group of the previous class. Then we will need to specify the objects and the actions of the group. We can use lists to hold the objects. . .

```
# group from 1.1 has as objects arrangements of a
# penny and a dime
arrangement1 = ['penny', 'nickel']
arrangement2 = ['nickel', 'penny']
objects = [arrangement1, arrangement2]
```

A Simple Group in Python

...and functions to represent the actions:

```
# our action takes arrangement1 and turns it into
# arrangement2, and
# vice-versa
def group_action(arrangement):
    if arrangement[0] == 'penny':
        return arrangement2
    else:
        return arrangement1

# check that this does what we want it to
for object in objects:
    print(group_action(object))
```

['nickel', 'penny']

['penny', 'nickel']

2026-06-22

CS Notes (Groups)

└ Creating Groups in Python

└ A Simple Group in Python

A Simple Group in Python

```
...and functions to represent the actions:
# our action takes arrangement1 and turns it into
# arrangement2, and
# vice-versa
def group_action(arrangement):
    if arrangement[0] == 'penny':
        return arrangement2
    else:
        return arrangement1

# check that this does what we want it to
for object in objects:
    print(group_action(object))

['nickel', 'penny']
['penny', 'nickel']
```

Example (Combining group actions using Python)

Find the result of performing the group action twice in a row using the code above.

Example (Words group again)

Implement the "three letter words" group in Python. Be careful to specify the objects and the actions!

2026-06-22

CS Notes (Groups)

└ Creating Groups in Python

└ Group Actions in Python

Example (Combining group actions using Python)

Find the result of performing the group action twice in a row using the code above.

Example (Words group again)

Implement the "three letter words" group in Python. Be careful to specify the objects and the actions!

There should be six objects and six actions that act on the objects!

Recall that a symmetry group of a geometric figure is the set of actions that preserves the shape of that figure.

Example (Symmetry group of equilateral triangle)

Implement the symmetry group of the equilateral triangle in Python.

2026-06-22

CS Notes (Groups)

└ Creating Groups in Python

└ Symmetry Groups in Python

This is more complicated than previous examples. Try working out examples by hand, then implementing in Python with a `for` loop.

Recall that a symmetry group of a geometric figure is the set of actions that preserves the shape of that figure.

Example (Symmetry group of equilateral triangle)

Implement the symmetry group of the equilateral triangle in Python.